A Paley–II analogue of the Scarpis–Đoković construction for prime powers $q \equiv 1 \pmod 4$

Jeffery Kline

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Abstract

A Hadamard matrix of order m is a $\{1, -1\}$-matrix H with $HH^T = mI_m$. Scarpis (1898) showed that for a prime $p \equiv 3 \pmod 4$ a Hadamard matrix of order $p+1$ can be lifted to one of order $p(p+1)$, and Đoković extended the lift to all prime powers $q \equiv 3 \pmod 4$, asking for the analogous procedure for $q \equiv 1 \pmod 4$. That case was settled by Farouk and Wang (2020), who lift an input Hadamard matrix of order $2(q+1)$, subject to eligibility conditions, to one of order $2q(q+1)$, verifying orthogonality by row-pair counting arguments in which the key inner products are imported from the Hadamard property of the input matrix. This note contributes a different method, not a new existence result. Both routes are constructive; the difference is procedure versus formula: their lift takes choices — an input matrix and a bijection — that can change the output, while the matrix here is given by a choice-free closed form, one matrix for each q. There is no input matrix: the finite bands are given entrywise in closed form by the quadratic character on $\mathbb{F}_q$ composed with the 2×2 Paley–II replacement blocks, and the Hadamard property is proved by four block-Gram identities computed directly from character sums. The printed outputs differ as well: at $q = 5$ our matrix is not Hadamard-equivalent to the order-60 example printed by Farouk and Wang, although it need not differ from every output of their procedure (Remark 1).

We then ask what the construction actually requires. Replacing the character by a function $\delta \colon G \to \{0, \pm 1\}$ with $\delta(0) = 0$ on a finite abelian group G of order n, and the shift by an arbitrary array T, the resulting matrix is Hadamard *if and only if* δ is of Paley type and T is a $(G, n; 1)$-difference matrix. The sufficiency half is known in greater generality, in the Butson setting and without commutativity, from Seberry and from Nuñez Ponasso; what is new here is the converse, that the two conditions are forced. They separate the sign pattern from the shift array and account for the shape of the family: the first makes $\{z : \delta(z) = 1\}$ a regular partial difference set with the Paley parameters, forcing $n \equiv 1 \pmod 4$; the second yields $n-1$ mutually orthogonal Latin squares, so an instance on a group of non-prime-power order would produce a projective plane of non-prime-power order.

A second reading of the same matrix is arithmetic rather than combinatorial. The Paley–II blocks are the real 2×2 representation of the Gaussian units, and through it the construction factors through a Butson matrix $\Gamma \in BH\big(q(q+1), 4\big)$ of half the order whose Turyn double is H. This locates the note against a second line in the literature: composing the complex Scarpis lift of Sargent, Lee and Rushall (2024) with the same doubling gives a further route to the existence statement, alongside Farouk and Wang, and Γ is equivalent to one of their outputs.

We also make the family's coverage explicit: every order $2q(q+1)$ whose odd part is at most 3000 is already recorded in the standard tables, and the smallest order lying beyond that audited range, $q = 81$ (order $13284 = 4 \cdot 41 \cdot 81$), also follows from a classical Turyn product using T-matrices of order 41 and Williamson-type matrices of order $81 = 9^2$. We exhibit the witnessing constructions, identify the smallest unaudited order, give a second worked example lying beyond the reach of the elementary screen, and tabulate, for $q \le 3000$, those orders for

which a deliberately restricted elementary screen supplies no witness, emphasising that these are *unaudited* rather than open.

1 Introduction

For a prime power q let $\mathbb{F} = \mathbb{F}_q$ and let $\chi\colon \mathbb{F} \to \{0, \pm 1\}$ be the quadratic character, $\chi(0) = 0$, $\chi(x) = 1$ for nonzero squares and -1 otherwise. Paley's two constructions split on the residue of q: for $q \equiv 3 \pmod 4$ Paley I gives a Hadamard matrix of order $q + 1$, and for $q \equiv 1 \pmod 4$ Paley II gives one of order $2(q + 1)$ [4].

Scarpis [5] showed that for a prime $p \equiv 3 \pmod 4$ a Hadamard matrix of order $p+1$ lifts to one of order $p(p+1)$. Đoković [11] replaced the prime by an arbitrary prime power: if $q \equiv 3 \pmod 4$ is a prime power there is a Hadamard matrix of order $q(q+1)$; his note closes with the open problem of an analogous procedure for $q \equiv 1 \pmod 4$.

Because the Paley–II seed has order $2(q + 1)$, the natural Scarpis-type target in the $q \equiv 1 \pmod 4$ case is

$$q \cdot 2(q + 1) = 2q(q + 1).$$

We construct such a matrix directly from χ. Throughout, $q \equiv 1 \pmod 4$ is a prime power, so $\chi(-1) = 1$; this is the only place the residue class is used, and it is exactly what fails for $q \equiv 3 \pmod 4$.

The existence of Hadamard matrices of order $2q(q+1)$ for prime powers $q \equiv 1 \pmod 4$ is due to Farouk and Wang [1]; the contribution of this note is the method of construction and proof. Farouk and Wang obtain the matrix as the output of a lifting procedure applied to an input Hadamard matrix of order $2(q + 1)$ meeting certain eligibility conditions, and their verification proceeds row pair by row pair, by counting arguments in which the key inner-product values are imported from the Hadamard property of the input matrix. Here there is no input matrix: the finite bands are written entrywise in the closed form $\psi(\chi(y - x - t))$, and the Hadamard property follows from four block-Gram identities whose ingredients are computed directly from the quadratic character — the sums (2.1) and $\sum_z \chi(z(z - d)) = -1$, together with the block identity $P + ZR = 0$. The proof is thus self-contained at the level of character sums, and the single place where the residue class enters, $\chi(-1) = 1$, is isolated. A secondary contribution is the coverage analysis of Appendix A, which has no counterpart in [1]: it locates the initial range against audited tables of known Hadamard orders and applies a restricted elementary screen beyond that range.

2 The Paley–II blocks

Put

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and let $\psi\colon \{0, \pm 1\} \to \{\pm 1\}^{2\times 2}$ be $\psi(0) = Z$, $\psi(1) = P$, $\psi(-1) = -P$; these are the Paley–II replacement blocks, and $ZR = -P$. Write $\mathcal{R} = J_q \otimes I_2$, where J_q is the $q \times q$ all-ones matrix. We use repeatedly

$$\sum_{u\in\mathbb{F}} \psi(\chi(u)) = Z, \tag{2.1}$$

which holds because the nonzero squares and non-squares are equinumerous, and the character sum

$$\sum_{z\in\mathbb{F}} \chi\big(z(z - d)\big) = -1 \qquad (d \neq 0). \tag{2.2}$$

For the latter, the term $z=0$ contributes $\chi(0)=0$, while for $z\neq 0$ we have $z(z-d)=z^2\big(1-dz^{-1}\big)$ with $\chi(z^2)=1$; as z runs over $\mathbb{F}^\times$ the element $u=1-dz^{-1}$ runs over $\mathbb{F}\setminus\{1\}$, so the sum equals $\sum_{u\neq 1}\chi(u)=-\chi(1)=-1$ because $\sum_{u\in\mathbb{F}}\chi(u)=0$.

3 The construction

For $t\in\mathbb{F}$ let K_t be the $2q\times 2q$ matrix with rows and columns indexed by pairs (x,a), $x\in\mathbb{F}$, $a\in\{0,1\}$, whose 2×2 block in position (x,y) is

$$K_t[x,y]=\psi\big(\chi(y-x-t)\big).$$

Lemma 1. *For every $t\in\mathbb{F}$, $K_tK_t^T=2(q+1)I_{2q}-2\mathcal{R}$.*

Proof. The (x,x') block of $K_tK_t^T$, with $d=x'-x$ and $z=y-x-t$, equals $\sum_z\psi(\chi(z))\psi(\chi(z-d))^T$. For $d=0$ each summand is $2I_2$, giving $2qI_2$. For $d\neq 0$ the terms $z=0$ and $z=d$ carry opposite skew parts and cancel, using $\chi(-d)=\chi(d)$ (here $\chi(-1)=1$); for $z\neq 0,d$, $\psi(\chi(z))\psi(\chi(z-d))^T=2\chi(z)\chi(z-d)I_2$, and (2.2) yields $-2I_2$. These are the diagonal and off-diagonal blocks of $2(q+1)I-2\mathcal{R}$. ☐

Lemma 2. *For $r,s\in\mathbb{F}$,*

$$\sum_{a\in\mathbb{F}}K_{ar}K_{as}^T=\begin{cases}2q(q+1)I_{2q}-2q\mathcal{R}, & r=s,\\ 2\mathcal{R}, & r\neq s.\end{cases}$$

Proof. The case $r=s$ is q copies of Lemma 1. For $r\neq s$, in the (x,x') block put $z=y-x-ar$; the second argument becomes $z-(x'-x)+a(r-s)$, which ranges over all of $\mathbb{F}$ independently of z as a does. By (2.1) each block is $ZZ^T=2I_2$, i.e. the corresponding block of $2\mathcal{R}$. ☐

For $r\in\mathbb{F}$ let B_r be the $2q\times 2q$ matrix whose row (x,a) equals row (r,a) of K_0 for every x; thus B_r has only two distinct rows. From Lemma 1, $B_rB_s^T=2q\mathcal{R}$ if $r=s$ and $-2\mathcal{R}$ if $r\neq s$.

Form the $2q\times 2q(q+1)$ block rows $M_r=\big[\,B_r\mid K_{ar}\ (a\in\mathbb{F})\,\big]$.

Lemma 3. *$M_rM_s^T=2q(q+1)I_{2q}$ if $r=s$, and 0 if $r\neq s$.*

Proof. $M_rM_s^T=B_rB_s^T+\sum_a K_{ar}K_{as}^T$. For $r=s$ the $2q\mathcal{R}$ of the border cancels the $-2q\mathcal{R}$ of Lemma 2; for $r\neq s$ the $-2\mathcal{R}$ cancels the $2\mathcal{R}$. ☐

Let C be the Paley conference matrix indexed by $\{\infty\}\cup\mathbb{F}$, namely $C_{u,u}=0$, $C_{\infty,x}=C_{x,\infty}=1$, $C_{x,y}=\chi(y-x)$, and let S be the order-$2(q+1)$ Paley–II Hadamard matrix obtained by replacing $0,1,-1$ by $Z,P,-P$. Delete the two rows of S indexed by ∞; in each remaining row repeat the ∞-column pair q times unchanged, and replace each finite column pair by its product with R, repeated q times. This is a $2q\times 2q(q+1)$ matrix M_∞. In both cases the q copies of a given column pair are placed *consecutively*, so that a column pair V (of size $2q\times 2$) becomes the $2q\times 2q$ block $J_{1,q}\otimes V=[\,V\mid V\mid\cdots\mid V\,]$. Thus M_∞ carries the same column partition as M_r above, an initial $2q$-wide ∞-block followed by q blocks of width $2q$ indexed by $\mathbb{F}$, and for each $a\in\mathbb{F}$ the a-block of M_∞ occupies the columns that K_{ar} occupies in M_r. This alignment is what the block-by-block summation in Lemma 4 uses; the interleaved reading, in which the copies of distinct column pairs alternate, does not yield a Hadamard matrix.

Lemma 4. *$M_\infty M_\infty^T=2q(q+1)I_{2q}$ and $M_\infty M_r^T=0$ for all $r\in\mathbb{F}$.*

Proof. The first identity holds because S is Hadamard [4], each column pair is repeated q times, and right multiplication by R preserves row inner products. For the cross identity, summing a cap row against a row of M_r and using (2.1) on every $2q$-block leaves $\big(P_\alpha + (\sum_a S[e,a]_\alpha)R\big) \cdot Z_\beta$; the inner sum of finite entries of a row of S is Z_α, and $P + ZR = P - P = 0$. $\square$

Theorem 1. *Let $q \equiv 1 \pmod 4$ be a prime power. Then*

$$H = \begin{bmatrix} M_\infty \\ M_r \ (r \in \mathbb{F}) \end{bmatrix}$$

is a Hadamard matrix of order $2q(q+1)$.

Proof. H is $\{1, -1\}$ and square of order $2q + q \cdot 2q = 2q(q+1)$. Lemmas 3 and 4 give $HH^T = 2q(q+1)I$. $\square$

Corollary 1. *For every prime power $q \equiv 1 \pmod 4$ there is a Hadamard matrix of order $2q(q+1)$; the smallest non-prime instance is $q = 9$, of order 180.*

The construction has been verified by full Gram computation for all $q \le 49$ (orders $60, \ldots, 4900$, including the extension fields $q = 9, 25, 49$) and by streamed row-orthogonality checks for $q = 81$ and $q = 125$ (orders 13284 and 31500).

4 Relation to the Farouk–Wang construction

Theorem 1 gives, for the $q \equiv 1 \pmod 4$ case, an explicit form of the Scarpis-type family that Đoković [11] asked for. That family was already constructed by Farouk and Wang [1, 2] from the same order-$2(q+1)$ Paley–II seed. This section records how the two routes differ and where the orders of the family sit relative to the audited tables of known Hadamard orders.

While Theorem 1 and [1, Cor. 2.1] assert the same existence result, the routes differ. Farouk and Wang obtain the order-$2q(q+1)$ matrix from a lifting procedure applied to a Hadamard matrix of order $2(q+1)$ satisfying the eligibility conditions of [1, Lem. 2.2] — a formally larger input class, with Paley II the only exhibited member — and verify orthogonality row pair by row pair through counting arguments (row sums, balanced sign-pair tallies, and a unique-intersection computation), importing the key inner-product values from the Hadamard property of the input matrix. The present note fixes the Paley–II seed and writes the finite bands entrywise in the closed form $\psi(\chi(y - x - t))$; the Hadamard property follows from four block-Gram identities in which the corresponding quantities are computed directly from the character sums (2.1) and $\sum_z \chi(z(z-d)) = -1$, together with $P + ZR = 0$ and the Hadamard property of the Paley–II seed S (itself a character-sum identity). The correspondence is exact: their unique-intersection index matches the affine collision $x + ar = x' + as$ in the substitution step of Lemma 2, and their imported inner products match the values that Lemma 1 derives from $\sum_z \chi(z(z-d)) = -1$. The devices used for the cap band also differ: a column permutation arranging balanced sign pairs in [1] versus right multiplication by R here.

Lemma 5. *Let $M = (m_{ic})$ have all entries of modulus 1. The multiset of $\left|\sum_c m_{ic}\overline{m_{jc}}m_{kc}\overline{m_{lc}}\right|$, taken over quadruples of distinct rows — equivalently, the multiset of its squares — is unchanged by permuting rows or columns, by multiplying any row or column by a scalar of modulus 1, and by complex conjugation.*

Proof. Permuting rows permutes the quadruples and permuting columns permutes the summands. Multiplying row i by u multiplies every summand by u or by $\bar{u}$, and multiplying column c by v multiplies the c-th summand by $v\bar{v}v\bar{v} = |v|^4 = 1$; conjugating M conjugates every sum. No absolute value changes in any case. □

For a $\{1, -1\}$-matrix the conjugations are vacuous and the admissible scalars are ± 1, which is the form used below and in Remark 4; the column profile of M is the row profile of M^T. The two constructions do not merely present the same matrix differently: at $q = 5$ the matrix of Theorem 1 is not Hadamard-equivalent to the order-60 example printed in the appendix of [1] — the multiset of absolute row-quadruple correlations $\left|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}\right|$, an invariant of Hadamard equivalence, differs for the two matrices (Figure 1). Since Farouk and Wang note that their output may vary with the choice of bijection α, the matrix of Theorem 1 may still be equivalent to some other output of their procedure; see Remark 1.

Remark 1 (unrefereed computational report)**.** *The following is* not *a claim of this note. It was produced by automated agents, has been checked here only by re-running the computation, and has not been refereed; we record it because it bears directly on the paragraph above. The report is that at* $q = 5$ *the matrix of Theorem 1* is *Hadamard-equivalent to an output of the Farouk–Wang procedure — to* $\Psi_{5,\alpha}(P_5')$ *for* α *any of the twenty affine bijections* $k \mapsto ak+b$ *of* $\mathbb{F}_5$ *— by an explicit signed row and column permutation, exhibited and checked by exact integer identity in* `code/alpha_family.py`*. If it is right, the inequivalence recorded above separates the matrix of Theorem 1 from the particular representative printed in [1], not from the family* $\Psi_{5,\alpha}$*; the same computation reports that three of the four profiles arising in that family do differ from ours, so the matrix is equivalent to some outputs and inequivalent to others. Two caveats are essential. It covers* $q = 5$ *alone. And it rests on a reading of [1, Steps 4–7] that required correcting the permutation displayed in the proof of their Lemma 2.2(2), which at* $q = 5$ *does not satisfy their own condition; their Theorem 2.2 and Corollary 2.1 are unaffected. Independent verification is wanted.*

Since $q \equiv 1 \pmod 4$, the 2-adic valuation of the order is exactly 2:

$$2q(q+1) = 4 \cdot q \cdot \tfrac{q+1}{2}, \qquad q \text{ and } \tfrac{q+1}{2} \text{ both odd.} \tag{4.1}$$

Thus every order is $4 \times (\text{odd})$, so it is never a Kronecker product of two smaller Hadamard matrices, and its odd part $q(q+1)/2$ is always composite. That last fact is a structural description and not an exclusion: of the 195 odd $m \le 2999$ for which [14] records no Hadamard matrix of order $4m$, fully 71 are composite, so compositeness of the odd part does not by itself keep an order off the open list. Whether the family meets that list has to be checked value by value, and Appendix A.1 does so for the whole range $q \le 73$. A second and more useful consequence is that (4.1) has the form needed for Turyn's construction: when T-matrices of order t and Williamson-type matrices of order w are available, one obtains a Hadamard matrix of order $4tw$ [7, 8], and here $\{t, w\} = \{q, (q+1)/2\}$. The coverage analysis of Appendix A makes both points quantitative: all orders with odd part ≤ 3000 are recorded in the standard tables, the first order beyond that range is covered by such a Turyn product, and a restricted screen beyond the audited range records exactly which orders it does and does not cover. Failure of that screen is not a claim that an order is open.

5 What the construction needs

Theorem 1 is stated over $\mathbb{F}_q$, but the field enters the proofs of Lemmas 1–4 in only two ways: through the character sums attached to χ, and, at exactly one line in the proof of Lemma 2, through the

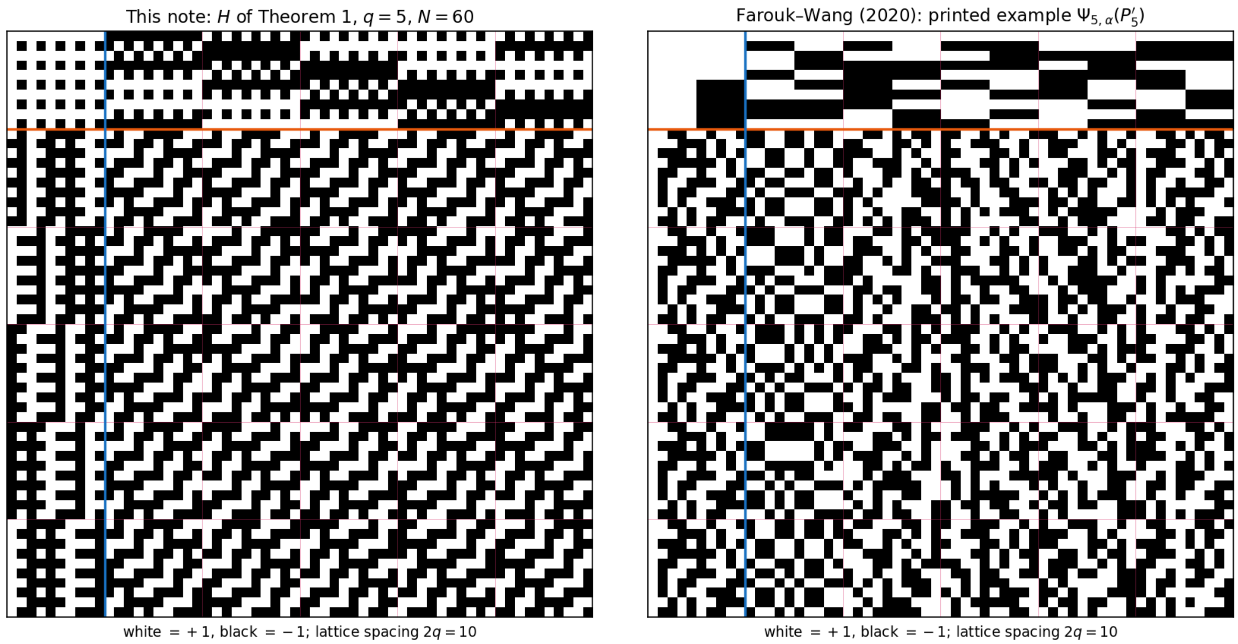

Figure 1: The two constructions at $q = 5$ (order 60; white $= +1$, black $= -1$; lattice spacing $2q = 10$). Left: the matrix of Theorem 1 — the cap band M_∞ above the orange separator is a family of period-two column patterns, and the finite bands M_r carry the diagonal texture of the shifts $\psi(\chi(y - x - ar))$. Right: the order-60 example $\Psi_{5,\alpha}(P_5')$ printed in the appendix of [1], whose top band is a coarse tiling (its border block consists of four constant 5×5 quadrants) and whose finite bands show no diagonal structure. The two matrices are Hadamard-inequivalent: over the $\binom{60}{4} = 487{,}635$ row quadruples, the right matrix attains $\left|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}\right| = 28$ on 400 quadruples, the left on none, and this multiset is invariant under row and column permutations and negations (and the comparison fails in both orientations, covering the transpose convention as well). This separates the printed representative only: outputs of the Farouk–Wang procedure for other choices of α are not addressed.

fact that $a \mapsto a(r - s)$ is a bijection of $\mathbb{F}$ when $r \neq s$. This section replaces each of the two by an abstract hypothesis and shows that the resulting pair of conditions is not merely sufficient but necessary. The conclusion is a clean division of labour — one condition on the sign pattern, one on the shift array — and, through the second condition, an explanation of why the family is confined to prime powers.

Let G be a finite abelian group, written additively, with $|G| = n$. Replace χ by a function $\delta\colon G \to \{0, \pm 1\}$ with $\delta(0) = 0$, and replace the shift ar by an arbitrary array $T\colon G \times G \to G$. Put

$$K_t[x, y] = \psi\big(\delta(y - x - t)\big) \qquad (x, y \in G),$$

let B_r be the $2n \times 2n$ matrix each of whose rows (x, α) equals row (r, α) of K_0, and set

$$M_r = \big[\, B_r \mid K_{T(r,a)}\ (a \in G) \,\big], \qquad r \in G.$$

Let C be indexed by $\{\infty\} \cup G$ with $C_{u,u} = 0$, $C_{\infty,x} = C_{x,\infty} = 1$, $C_{x,y} = \delta(y - x)$, let S be its ψ-replacement, and let M_∞ be built from S exactly as in §3. Write

$$H(G, \delta, T) = \begin{bmatrix} M_\infty \\ M_r\ (r \in G) \end{bmatrix},$$

a $\{1,-1\}$-matrix of order $2n(n+1)$. Taking $G=(\mathbb{F}_q,+)$, $\delta=\chi$ and $T(r,a)=ar$ returns the matrix of Theorem 1 entry for entry.

Call δ *of Paley type* if $\delta(0)=0$, $\delta(z)\neq 0$ for $z\neq 0$, $\delta(-z)=\delta(z)$, and $\sum_{z\in G}\delta(z)\delta(z-d)=-1$ for every $d\neq 0$. Call T a $(G,n;1)$-*difference matrix* if for all $r\neq s$ the map $a\mapsto T(r,a)-T(s,a)$ is a bijection of G. Such a T attains Jungnickel's bound $m\le\lambda|G|$ for (G,m,λ)-difference matrices [21, Thm. 2.2], and is in the standard terminology a *generalized Hadamard matrix* $GH(n,1)$ over G.

Lemma 6. *For $c\in\{0,\pm1\}$ put $e(c)=1-c^2$. Then*

$$\psi(c)\psi(c')^T=2\big[cc'+e(c)e(c')\big]I_2+2\big[c\,e(c')-c'e(c)\big]R\qquad(c,c'\in\{0,\pm1\}),$$

and consequently, for every $\delta\colon G\to\{0,\pm1\}$ and $c\in G$,

$$B(c):=\sum_{z\in G}\psi(\delta(z))\psi(\delta(z-c))^T=2\big[A(c)+E(c)\big]I_2+2W(c)R,\tag{5.1}$$

where $A(c)=\sum_z\delta(z)\delta(z-c)$, $E(c)=\#\{z:\delta(z)=\delta(z-c)=0\}$ and $W(c)=\sum_{w:\,\delta(w)=0}\big[\delta(w+c)-\delta(w-c)\big]$.

Proof. The first identity is a check on nine pairs, and reduces to $PP^T=ZZ^T=2I_2$, $(cP)(c'P)^T=2cc'I_2$ for $c,c'\neq 0$, $PZ^T=2R$ and $ZP^T=-2R$. Summing it over $z\in G$ and collecting the four resulting sums gives (5.1); in the skew part the substitution $z-c=w$ turns $\sum_z\delta(z)e(\delta(z-c))$ into $\sum_{\delta(w)=0}\delta(w+c)$ and $\sum_z\delta(z-c)e(\delta(z))$ into $\sum_{\delta(w)=0}\delta(w-c)$. □

Theorem 2. *Let G be a finite abelian group of order n, let $\delta\colon G\to\{0,\pm1\}$ satisfy $\delta(0)=0$, and let $T\colon G\times G\to G$. Then*

$$H(G,\delta,T)\,H(G,\delta,T)^T=2n(n+1)\,I$$

if and only if δ is of Paley type and T is a $(G,n;1)$-difference matrix.

The sufficiency direction is not new, and in the Butson setting it is already known in greater generality. Seberry [15], building on Rajkundlia, gave a generalized-Hadamard form of the Scarpis construction, and Nuñez Ponasso [16, Thm. 4.3.3] proves that a $BH(n+1,m)$ together with a $GH(n,G)$ with $|G|=n$ yields a $BH(n(n+1),m)$, for any group G and any m. Taking $m=4$ and composing with the doubling of §6 recovers the "if" half of Theorem 2 without the hypothesis that G be abelian. What is proved here and does not follow from those results is the converse: that the two conditions are *forced*, so that the construction admits no other input. That is the direction Corollaries 2 and 3 rest on.

Proof. Let B be as in (5.1) and put $\mu_{rs}(u)=\#\{a\in G: T(r,a)-T(s,a)=u\}$. Rerunning the block computations of §3 with χ replaced by δ and ar by $T(r,a)$ — no step of which uses anything beyond the abelian group law — gives every Gram block of H. For $r,s\in G$ and $x,x'\in G$, with $d=x'-x$,

$$\big(M_rM_s^T\big)[x,x']=B(s-r)+\sum_{u\in G}\mu_{rs}(u)\,B(d-u),\tag{5.2}$$

the first term from $B_rB_s^T$ and the second from $\sum_a K_{T(r,a)}K_{T(s,a)}^T$ after the substitutions $z=y-x-T(r,a)$, $u=T(r,a)-T(s,a)$. Likewise $\big(M_\infty M_\infty^T\big)[e,e']=2nI_2+n\,B(e'-e)$, because the ∞-column pair contributes $nPP^T=2nI_2$ and $RR^T=I_2$ makes the finite columns contribute $nB(e'-e)$.

Finally, writing $\Sigma = \sum_{c\in G}\psi(\delta(c)) = n_0 Z + \sigma P$ with $n_0 = \#\{z : \delta(z) = 0\}$ and $\sigma = \sum_z \delta(z)$, every block of $M_\infty M_r^T$ equals

$$P\Sigma^T + \Sigma R \Sigma^T = 2\sigma I_2 + 2n_0 R - 2\big(n_0^2 + \sigma^2\big)R, \tag{5.3}$$

using $PP = 2I_2$, $PZ = 2R$, $ZZ = 2I_2$, $ZP = -2R$, $ZR = -P$, $PR = Z$ and $RZ = P$; the term $P\Sigma^T$ comes from the border and $\Sigma R\Sigma^T$ from the n finite column blocks.

Necessity. Take $r = s$ in (5.2): then $\mu_{rr}(u) = n$ if $u = 0$ and 0 otherwise, so the block is $2nI_2 + nB(d)$. Vanishing off the diagonal forces

$$B(d) = -2I_2 \qquad \text{for all } d \neq 0,$$

i.e. $A(d) + E(d) = -1$ and $W(d) = 0$ for $d \neq 0$. Next, (5.3) vanishes only if $\sigma = 0$ and $n_0 = n_0^2 + \sigma^2$, hence $n_0 \in \{0, 1\}$; since $\delta(0) = 0$ we get $n_0 = 1$, so δ vanishes only at 0. Then $E(d) = 0$ for $d \neq 0$, so $A(d) = -1$, and $W(d) = \delta(d) - \delta(-d) = 0$, so δ is even: δ is of Paley type. Finally take $r \neq s$ in (5.2). Since $B(0) = 2nI_2$ and $B(c) = -2I_2$ for $c \neq 0$,

$$\big(M_r M_s^T\big)[x, x'] = -2I_2 + 2\big[(n+1)\mu_{rs}(d) - n\big]I_2 = 2(n+1)\big[\mu_{rs}(d) - 1\big]I_2,$$

and as $d = x' - x$ runs over all of G, vanishing forces $\mu_{rs} \equiv 1$: T is a $(G, n; 1)$-difference matrix.

Sufficiency. If δ is of Paley type then $n_0 = 1$ and $\sigma^2 = \sum_d A(d) = (n-1) - (n-1) = 0$, so (5.3) vanishes and $M_\infty M_r^T = 0$; also $B(0) = 2nI_2$ and $B(d) = -2I_2$ for $d \neq 0$, whence $M_\infty M_\infty^T = 2n(n+1)I$ and, in (5.2) with $r = s$, $M_r M_r^T = 2n(n+1)I$. If in addition $\mu_{rs} \equiv 1$ for $r \neq s$, the display above gives $M_r M_s^T = 0$. Hence $HH^T = 2n(n+1)I$. □

Corollary 2. *If δ is of Paley type on G, $|G| = n$, then $\sum_z \delta(z) = 0$, the set $D = \{z : \delta(z) = 1\}$ has $(n-1)/2$ elements, and in the group ring $\mathbb{Z}[G]$*

$$\widehat{D}^{\,2} + \widehat{D} = \tfrac{n-1}{4}\big(1 + \widehat{G}\big),$$

so that D is a regular partial difference set with the Paley parameters $\big(n, \frac{n-1}{2}, \frac{n-5}{4}, \frac{n-1}{4}\big)$. In particular $n \equiv 1 \pmod 4$.

Proof. In $\mathbb{Z}[G]$, $\delta = 2\widehat{D} - \widehat{G} + 1$ and, δ being even, $\delta^2 = \sum_d A(d)\, d = (n-1)\cdot 1 - (\widehat{G} - 1) = n \cdot 1 - \widehat{G}$; in particular $\sigma^2 = \sum_d A(d) = 0$, so $|D| = (n-1)/2$. Expanding $(2\widehat{D} - \widehat{G} + 1)^2$ with $\widehat{D}\widehat{G} = |D|\widehat{G}$ and $\widehat{G}^2 = n\widehat{G}$ and comparing with $n \cdot 1 - \widehat{G}$ gives the displayed identity. Reading off coefficients gives $\lambda = (n-5)/4$ on D and $\mu = (n-1)/4$ off $D \cup \{0\}$; integrality forces $n \equiv 1 \pmod 4$. □

Corollary 3. *Let G be abelian of order n and let T be a $(G, n; 1)$-difference matrix. Then there exist $n-1$ mutually orthogonal Latin squares of order n, hence a projective plane of order n. Consequently, by Theorem 2, an instance of the construction on a group of non-prime-power order would produce a projective plane of non-prime-power order.*

Proof. Put $\widetilde{T}(r, a) = T(r, a) - T(0, a)$; this is again a difference matrix and $\widetilde{T}(0, a) = 0$, so $a \mapsto \widetilde{T}(r, a)$ is a bijection for every $r \neq 0$. For $r \neq 0$ define $L_r[x, a] = x + \widetilde{T}(r, a)$. Each row and each column of L_r is a bijection, so L_r is a Latin square on G. For $r \neq s$ the pair $\big(L_r[x, a], L_s[x, a]\big)$ determines $\widetilde{T}(r, a) - \widetilde{T}(s, a)$, hence a, hence x; so L_r and L_s are orthogonal. (The equivalence between $(G, n; 1)$-difference matrices and G-regular sets of $n-1$ mutually orthogonal Latin squares of order n is due to Jungnickel [22, Thm. 1]; the short proof of the direction needed here is included for self-containedness.) These $n-1$ squares are a complete set, and a complete set of $n-1$ mutually orthogonal Latin squares of order n is equivalent to a projective plane of order n [20]. The last sentence is Theorem 2 applied in the direction "Hadamard $\Rightarrow$ T is a difference matrix". □

Remark 2. *The two slots of Theorem 2 have very different supply. By Corollary 2 the δ-slot asks exactly for a Paley type partial difference set, and these are classified: Wang [17] proved that a Paley type partial difference set in an abelian group of non-prime-power order forces $|G| = m^4$ or $9m^4$ with $m > 1$ odd, and Polhill [19] had constructed them at all such orders — the first construction at a non-prime-power order, in $\mathbb{Z}_3^2 \times \mathbb{Z}_p^{4t}$, being [18] — so [17, Thm. 1.2] an odd $v > 1$ admits a Paley type partial difference set in some abelian group if and only if v is a prime power $\equiv 1 \pmod 4$, or $v = m^4$ or $9m^4$ with $m > 1$ odd. The smallest non-prime-power in that list is $9 \cdot 5^4 = 5625$. So the δ-slot does reach beyond prime powers. The T-slot does not, as far as anything known: in every known example of a $GH(|G|, \lambda)$ over G the group order $|G|$ is a prime power, and if G is not elementary abelian then $|G|$ is a square [20, p. 303]; and by Corollary 3 an instance at $n = 5625$ would supply a projective plane of order $5625 = 3^2 \cdot 5^4$, and no projective plane of non-prime-power order is known. The Bruck–Ryser theorem [25] gives no help here, since every candidate order m^4 or $9m^4$ is a perfect square and hence a sum of two squares. The smallest order this construction could conceivably reach outside the prime powers is therefore $2 \cdot 5625 \cdot 5626 = 63\,292\,500$, and reaching it entails settling a much older question. For scale, plane order 10 was excluded by exhaustive computation [26], and re-established with machine-verifiable nonexistence certificates only in 2021 [24]; order 12 is still undecided (see e.g. [27]).*

Remark 3. *Farouk and Wang [2] abstract the shift array of a Scarpis-type construction in a related but formally weaker way: a pair of Latin squares L, L' of order n is* eligible *if for all i, i' there is a unique j with $L[i,j] = L'[i',j]$ [2, Def. 2.2]; at most $n-1$ squares can be pairwise eligible [2, Thm. 2.4]; the squares $x_i + bx_j$ over $\mathbb{F}_q$ realise the bound [2, Cor. 2.5]; and complete eligible sets need not be sets of mutually orthogonal Latin squares, which is why they raise the construction of a complete eligible set of non-prime-power order as an open problem. Corollary 3 says that this loophole is closed for a closed-form construction of the present shape. The layout of §3 produces only the group-developed squares $L_r[x,a] = x + \widetilde{T}(r,a)$, and for those three conditions coincide: eligibility of L_r, L_s says that for every $c \in G$ there is a unique a with $\widetilde{T}(r,a) - \widetilde{T}(s,a) = c$, which is the difference-matrix condition, which by the proof of Corollary 3 is orthogonality (and conversely: if L_r, L_s are orthogonal then for each $c \in G$ the n cells (x,a) with $L_r[x,a] - L_s[x,a] = c$ are in bijection with the n ordered pairs of values differing by c, so exactly one a satisfies $\widetilde{T}(r,a) - \widetilde{T}(s,a) = c$). The coincidence is an observation about this layout rather than new machinery: the underlying correspondence is Jungnickel's [22, Thm. 1], and group-developed Latin squares and their orthomorphisms are the subject of a standard monograph [23]. An eligible-but-not-orthogonal set of non-prime-power order, if one exists, cannot be group-developed, and so cannot be fed to a closed form of this kind.*

Remark 4. *Theorem 2 also shows that fixing q does not fix the matrix. At $q = 9$ the δ-slot is rigid: of the $2^4 = 16$ even sign patterns on $\mathbb{Z}_3^2$ vanishing at 0, exactly 6 are of Paley type, and all are equivalent to χ under $\mathrm{Aut}(\mathbb{Z}_3^2)$. The T-slot is not rigid. Let N be the Dickson nearfield of order 9: the additive group is again $\mathbb{Z}_3^2 = (\mathbb{F}_9, +)$, with $r \circ a = ra$ for r a square and $r \circ a = ra^3$ for r a non-square. Each $a \mapsto r \circ a$ is additive and $r \circ a - s \circ a$ vanishes only at $a = 0$ for $r \neq s$, so $T(r,a) = r \circ a$ is a $(\mathbb{Z}_3^2, 9; 1)$-difference matrix and Theorem 2 yields a second Hadamard matrix of order 180, obtained from Theorem 1 by the single substitution $\mathbb{F}_9 \rightsquigarrow N$. It is not Hadamard-equivalent to the matrix of Theorem 1: over the $\binom{180}{4} = 42\,296\,805$ row quadruples the multiset of $\left|\sum_c h_{ic}h_{jc}h_{kc}h_{lc}\right|$ attains the value 44 on $112\,752$ quadruples for the nearfield matrix and on none for the matrix of Theorem 1, and the comparison fails in all four row/column orientations. Since $\{a \mapsto r \circ a : r \in N\}$ is not closed under addition, the translation plane it coordinatises is not Desarguesian, and the matrix of Theorem 1 is the Desarguesian choice. We do not claim that the inequivalence is* caused *by non-Desarguesianness: relabelling the bands of the Desarguesian difference matrix already yields order-180 matrices inequivalent to that of Theorem 1. What the nearfield supplies is a canonical* *second choice, not the only source of inequivalence.*

6 The construction over the Gaussian integers

The blocks P, Z, R of §2 are not three unrelated objects: they are the real 2×2 representation of the Gaussian units, and through it Theorem 1 factors through a matrix of half the order over $\mathbb{Z}[i]$. This section makes that factorisation explicit and records what it implies about the relation of Theorem 1 to the literature. Nothing in §§1–4 depends on this section.

Let

$$\rho\colon \mathbb{Z}[i]\to M_2(\mathbb{Z}),\quad \rho(a+bi)=aI_2+bR,\qquad \varphi\colon \mathbb{Z}[i]\to M_2(\mathbb{Z}),\quad \varphi(w)=\rho(w)P.$$

Since $R^2=-I_2$, the map ρ is an injective ring homomorphism, and $\rho(w)^T=\rho(\bar{w})$. Directly from the definitions,

$$\varphi(1)=P,\qquad \varphi(-1)=-P,\qquad \varphi(-i)=-RP=Z,\qquad \varphi(i)=-Z,$$

the third identity being the relation $ZR=-P$ of §2 rewritten together with $PR=-RP$. Define $e\colon \mathbb{F}\to\{1,-1,-i\}\subset\mathbb{Z}[i]$ by

$$e(z)=\chi(z)\quad (z\neq 0),\qquad e(0)=-i. \tag{6.1}$$

Comparing with $\psi(0)=Z$, $\psi(\pm1)=\pm P$ gives

$$\psi(\chi(z))=\varphi\big(e(z)\big)\qquad \text{for every } z\in\mathbb{F}, \tag{6.2}$$

so the Paley–II replacement blocks are the Gaussian units in disguise, with e the quadratic character punctured by $-i$ rather than by 0. Since $PR=Z$, and since $\rho(\mathbb{Z}[i])$ is commutative (so that $\rho(-i)\rho(w)=\rho(w)\rho(-i)$; note P itself does *not* commute with R),

$$\varphi(w)R=\rho(w)PR=\rho(w)Z=\varphi(-iw), \tag{6.3}$$

so the right multiplication by R used to build the cap band M_∞ is multiplication of a Gaussian unit by $-i$.

Lemma 7. *For all $w,v\in\mathbb{Z}[i]$ we have $\varphi(w)\varphi(v)^T=2\rho(w\bar{v})$. Consequently, let $\Gamma=(\gamma_{uv})$ be an $n\times n$ matrix with entries in $\{1,-1,i,-i\}$ and let $\varphi(\Gamma)$ denote the $2n\times 2n$ $\{1,-1\}$-matrix whose (u,v) block is $\varphi(\gamma_{uv})$. Then the (u,u') block of $\varphi(\Gamma)\varphi(\Gamma)^T$ is $2\rho\big((\Gamma\Gamma^*)_{u,u'}\big)$, and $\varphi(\Gamma)$ is a Hadamard matrix of order $2n$ if and only if $\Gamma\Gamma^*=nI_n$.*

Proof. Using $PP^T=2I_2$ and $\rho(v)^T=\rho(\bar{v})$,

$$\varphi(w)\varphi(v)^T=\rho(w)PP^T\rho(v)^T=2\rho(w)\rho(\bar{v})=2\rho(w\bar{v}).$$

Summing over the common index, $\sum_v\varphi(\gamma_{uv})\varphi(\gamma_{u'v})^T=2\rho\big(\sum_v\gamma_{uv}\overline{\gamma_{u'v}}\big)=2\rho\big((\Gamma\Gamma^*)_{u,u'}\big)$. As ρ is injective and $\rho(n)=nI_2$, this block equals $2n\delta_{u,u'}I_2$ precisely when $(\Gamma\Gamma^*)_{u,u'}=n\delta_{u,u'}$. □

A matrix Γ with entries in $\{1,-1,i,-i\}$ and $\Gamma\Gamma^*=nI_n$ is a Butson–Hadamard matrix, written $\Gamma\in BH(n,4)$. Lemma 7 is thus the classical morphism $BH(n,4)\to BH(2n,2)$, obtained independently by Cohn [31] and Turyn [32]; we call $\varphi(\Gamma)$ the *Turyn double* of Γ.

Theorem 3. *Let $q \equiv 1 \pmod 4$ be a prime power and $n = q(q+1)$. Let Γ be the $n \times n$ matrix over $\mathbb{Z}[i]$ whose rows are indexed by $\mathbb{F}$ followed by $\mathbb{F} \times \mathbb{F}$, whose columns are indexed by $\mathbb{F}$ followed by $\mathbb{F} \times \mathbb{F}$, and whose entries are*

$$\Gamma\big[x;\, y\big] = 1, \qquad \Gamma\big[x;\, (a,y)\big] = -i\, e(a-x),$$
$$\Gamma\big[(r,x);\, y\big] = e(y-r), \qquad \Gamma\big[(r,x);\, (a,y)\big] = e(y-x-ar).$$

Then $\Gamma \in BH\big(q(q+1),4\big)$, and the matrix H of Theorem 1 is its Turyn double: $H = \varphi(\Gamma)$.

Proof. Read off the 2×2 blocks of H. In K_{ar} the (x,y) block is $\psi(\chi(y-x-ar)) = \varphi(e(y-x-ar))$ by (6.2); in the border block B_r the (x,y) block is $\psi(\chi(y-r)) = \varphi(e(y-r))$; in the cap band M_∞ the repeated ∞-column pair contributes $P = \varphi(1)$ and a finite column pair contributes $\psi(\chi(a-x))R = \varphi(e(a-x))R = \varphi(-i\, e(a-x))$ by (6.3). These are exactly the entries listed for Γ, and φ is injective on $\{1,-1,i,-i\}$, so $H = \varphi(\Gamma)$. By Theorem 1 and Lemma 7, $\Gamma\Gamma^* = nI_n$. □

Remark 5. *Lemma 7 converts the four block-Gram identities of §3 into scalar identities over $\mathbb{Z}[i]$; the 2×2 block algebra carries no information beyond ρ. Lemma 1, for instance, becomes*

$$\sum_{z\in\mathbb{F}} e(z)\overline{e(z-d)} = \begin{cases} q, & d = 0,\\ -1, & d \neq 0,\end{cases}$$

which is $\sum_z \chi(z(z-d)) = -1$ together with the cancellation $e(0)\overline{e(-d)} + e(d)\overline{e(0)} = -i\chi(d) + i\chi(d) = 0$ already used in the proof of Lemma 1 — the step where $\chi(-1) = 1$ enters. Likewise (2.1) becomes $\sum_{u\in\mathbb{F}} e(u) = -i$.

Remark 6. *Sargent, Lee and Rushall [33] carry the Scarpis lift over to complex Hadamard matrices: by their Corollary 1, for an odd prime power q a complex (Butson-type) Hadamard matrix of order $q+1$ yields one of order $q(q+1)$. Every entry of their output is a product of entries of the input and of its negated core, so a $BH(q+1,4)$ input gives a $BH(q(q+1),4)$ output. For $q \equiv 1 \pmod 4$ the Paley conference matrix C of order $q+1$ is symmetric with $CC^T = qI$, so $(C-iI)(C-iI)^* = CC^T + I = (q+1)I$ and $C - iI$ is such an input. (This is the standard conference-matrix construction $I \pm iC$, generally credited to [32]; the one-line verification is included because it is shorter than the citation.) Composing their Corollary 1 with Lemma 7 therefore yields a second route in the literature to the existence statement of Theorem 1, alongside [1]. The same move is available in Butson generality: composing [16, Thm. 4.3.3] on a $BH(q+1,4)$ seed with Lemma 7 — his Theorem 4.4.1 — is a third such route.*

The matrix Γ of Theorem 3 is in fact equivalent to one of their outputs. Index $C - iI$ by $\{\infty\} \cup \mathbb{F}$ with ∞ first and normalise it by dividing each row by its first entry and then each column by its first entry; the result N satisfies $N[\infty,u] = N[u,\infty] = 1$ and $N[x,a] = -i\, e(a-x)$ for $x, a \in \mathbb{F}$. Feeding N to their construction with the row indexed by ∞ deleted returns $\operatorname{diag}(I_q,\, iI_{q^2})\,\Gamma$; a diagonal unimodular row scaling lies in their equivalence group, so Γ is equivalent to that output. Over $\mathbb{Z}[i]$, then, the construction of §3 is the instance of theirs in which the seed, the deleted row and the bijection α are all determined by q; the distinction drawn in §4 against [1] — formula rather than procedure — is the accurate description of the difference here as well. The relation recorded here is, however, strictly closer than the one recorded in §4. There we note in addition that the two constructions yield inequivalent matrices; no corresponding statement is available here, and none is made. The matrix Γ is equivalent to an output of [33], not merely analogous to one.

Remark 7. Γ *is not the classical Butson matrix of its order. At $q = 5$, $\Gamma \in BH(30,4)$, and so is Turyn's $C_{30} + iI$ built from the Paley conference matrix of order* 30 *(that is, from the prime* 29*); the* *two are inequivalent. The multiset of* $\left|\sum_c \gamma_{ic}\overline{\gamma_{jc}}\gamma_{kc}\overline{\gamma_{lc}}\right|^2$ *over ordered quadruples of distinct rows is invariant under row and column permutations, multiplication of rows and columns by fourth roots of unity, and complex conjugation; it takes the value* 500 *on* 240 *quadruples for* Γ *and on none for* $C_{30} + iI$*, in both orientations.*

7 Open problems

Theorem 2 turns the construction into two independent slots, and most of what is not settled here is a question about one of them.

1. The T-slot as a program. For a fixed abelian G of order n, the admissible shift arrays are exactly the $(G, n; 1)$-difference matrices, equivalently the G-regular complete sets of $n - 1$ mutually orthogonal Latin squares [22, Thm. 1]. At $n = 9$ at least two are available — the field product $T(r, a) = ar$ and the Dickson nearfield product of Remark 4 — and they give Hadamard matrices of order 180 that are inequivalent. Remark 4 also shows that mere relabelling of the bands already produces inequivalent outputs, so the count of classes exceeds one for an uninteresting reason. The question is quantitative: as T ranges over all $(G, n; 1)$-difference matrices, how many Hadamard-equivalence classes of order $2n(n + 1)$ arise, and is the field choice distinguished among them by any invariant? Nothing here answers this even at $n = 9$.

2. Uniqueness in the δ-slot. By Corollary 2 a Paley-type δ is the same thing as a regular partial difference set with the Paley parameters. At $n = 9$ exactly 6 of the 16 admissible sign patterns are of Paley type and all 6 are equivalent under $\mathrm{Aut}(\mathbb{Z}_3^2)$, so the slot is rigid there. Is δ always unique up to $\mathrm{Aut}(G)$? The classification of Wang [17] settles which orders occur but not how many inequivalent sets occur at a given order.

3. Where the obstruction actually sits at non-prime-power order. Remark 2 locates the smallest conceivable non-prime-power instance at $n = 9 \cdot 5^4 = 5625$, order 63 292 500. The two slots are not symmetric there: the δ-slot *is* available, since Polhill [19] constructs Paley type partial difference sets at every order m^4 and $9m^4$, while no $(G, 5625; 1)$-difference matrix is known and one would supply a projective plane of order 5625. So for this construction the entire obstruction beyond the prime powers lies in the T-slot. Does a $(G, n; 1)$-difference matrix exist for any non-prime-power n?

4. The residue class $q \equiv 3 \pmod 4$. Đoković [11] extends Scarpis to prime powers there, producing order $q(q + 1)$ from a Paley I seed, and $\chi(-1) = -1$ is exactly what obstructs the argument of §3 — the residue class is used in one place only, the proof of Lemma 1. Is there a choice-free closed form in that class as well, and does it fall under a characterisation of the shape of Theorem 2?

5. Equivalence to the rest of the Farouk–Wang family. The inequivalence recorded in §4 is to the order-60 matrix printed in [1]. Their procedure varies with a bijection α, so whether the matrix of Theorem 1 is equivalent to *some* output of that procedure is open for $q > 5$; at $q = 5$ see the unrefereed report of Remark 1. Likewise for the complex lift of [33] beyond the equivalence noted in Remark 6.

A Coverage analysis

Write $N = 2q(q + 1)$ and $m = q(q + 1)/2$ for its odd part, so $N = 4m$. We assess coverage of N by prior constructions on four levels: the audited tables (§A.1), the first unaudited order (§A.2), a second worked example lying beyond the reach of the elementary test (§A.3), and that elementary-witness test itself, which isolates the candidate orders (§A.4).

A.1 Orders with odd part at most 3000

The surveys of Seberry and Yamada [8, 13] and the database of Cati and Pasechnik [14] record, for every odd $m \le 3000$, the least t for which a Hadamard matrix of order $2^t m$ is known. The default is $t = 2$, so that $N = 4m$ is known. Table 4 of [14] lists the 195 odd $m \le 2999$ with $t > 2$; of these 124 are prime and 71 are composite, the smallest entry being 167 and the smallest composite entry 515. Compositeness of m is therefore no guarantee on its own, and the check has to be made value by value. The twelve odd parts $m = q(q+1)/2$ arising here for $q \le 73$, namely

$$15,\ 45,\ 91,\ 153,\ 325,\ 435,\ 703,\ 861,\ 1225,\ 1431,\ 1891,\ 2701,$$

are each absent from that table; hence *every N with $m \le 3000$ is a known Hadamard order*. This is the range $q \le 73$, tabulated below with an explicit elementary witness where one exists (Paley I/II, or a Turyn product $4tw$).

q	$N = 2q(q+1)$	odd part m	witnessing construction
5	60	$3 \cdot 5$	Paley II, $p = 29$
9	180	$3^2 \cdot 5$	Paley II, $p = 89$
13	364	$7 \cdot 13$	Paley II, $p = 181$
17	612	$3^2 \cdot 17$	Turyn $4 \cdot 9 \cdot 17$
25	1300	$5^2 \cdot 13$	Turyn $4 \cdot 13 \cdot 25$
29	1740	$3 \cdot 5 \cdot 29$	Turyn $4 \cdot 15 \cdot 29$
37	2812	$19 \cdot 37$	Turyn $4 \cdot 19 \cdot 37$
41	3444	$3 \cdot 7 \cdot 41$	Paley II, $p = 1721$
49	4900	$5^2 \cdot 7^2$	Turyn $4 \cdot 25 \cdot 49$
53	5724	$3^3 \cdot 53$	Paley II, $p = 2861$
61	7564	$31 \cdot 61$	Turyn $4 \cdot 31 \cdot 61$
73	10804	$37 \cdot 73$	tables; no restricted witness (see §A.4)

The entry $q = 73$ is instructive: the restricted test below supplies no divisor-split witness for its odd part $2701 = 37 \cdot 73$, yet $N = 10804$ is a known Hadamard order because $m \le 3000$ is audited. Failure of the elementary test is therefore a property of that test, not of Hadamard existence.

A.2 The first unaudited order: $q = 81$

The smallest order of the family whose odd part exceeds 3000, and hence lies beyond the audited range, is

$$q = 81 = 3^4, \qquad N = 13284, \qquad m = 3321 = 3^4 \cdot 41 > 3000.$$

It is nevertheless a known Hadamard order. By (4.1),

$$13284 = 4 \cdot 41 \cdot 81 = 4tw, \qquad t = 41,\ w = 81,$$

and both ingredients are classical: T-matrices of order 41 come from the base sequences $BS(21, 20)$ [7, 12], and Williamson-type matrices of order $81 = 9^2$ come from Turyn's 9-power family [7, 8]. The Turyn array on these [8, 13] yields a Hadamard matrix of order $4 \cdot 41 \cdot 81 = 13284$. Thus $q = 81$ is the smallest unaudited order, and it is covered.

A.3 A second worked example: $q = 109$

The smallest order of the family for which the restricted test of §A.4 supplies no witness is

$$q = 109, \qquad N = 23980, \qquad m = 5995 = 5 \cdot 11 \cdot 109 > 3000.$$

It too is a known Hadamard order. By (4.1),

$$23980 = 4 \cdot 55 \cdot 109 = 4tw, \qquad t = 55,\ w = 109.$$

The first ingredient is already admitted by (C.1): $t = 55 \leq 59$ is T-known, the T-matrices of order 55 coming from the base sequences $BS(28, 27)$, which lie inside Đoković's classification of $BS(n+1, n)$ for $n \leq 30$ [29]; independently, [30, §2] records that T-sequences of every length $n \leq 100$ exist except possibly $n = 97$. Only the second ingredient is missing from (C.1), and it is supplied by a result of Miyamoto [28], in the reformulation of Seberry and Yamada [8, Cor. 8.8, part 1], as quoted by Đoković [30, Thm. 2.5]: *if* $q \equiv 1 \pmod 4$ *is a prime power, then there exist Williamson-type matrices of order* q *if there are Williamson-type matrices of order* $(q-1)/4$*, or an Hadamard matrix of order* $(q-1)/2$.

Here $q = 109$ is prime and $\equiv 1 \pmod 4$, and $(109 - 1)/4 = 27$. Williamson matrices of order 27 are known: $27 \leq 33$, so 27 is W-known in the sense of §A.4, and Seberry and Yamada record order 27 directly, their class w_1 of Williamson orders being $\{1, \ldots, 33, 37, 39, 41, 43\}$ [8, Table A.1 and legend, pp. 541–543]. No nonexistence result bears on 27: the smallest odd order at which Williamson matrices are known not to exist is 35, and $35 > 33$ (cf. Remark 8). Williamson matrices are in particular Williamson-type: symmetric circulants commute, so $AB^T = AB = BA = BA^T$, and $AA^T + BB^T + CC^T + DD^T = A^2 + B^2 + C^2 + D^2 = 4nI$. (The second hypothesis of the theorem as quoted cannot be used here: $(109 - 1)/2 = 54$ is neither 1 nor 2 nor divisible by 4, so no Hadamard matrix of order 54 exists. The conclusion is nonetheless over-determined rather than fragile: the companion part of the same corollary [8, Cor. 8.8(2), p. 505] asks only for a *symmetric conference* matrix of order $(q-1)/2 = 54$, which exists by Paley's construction since 53 is a prime $\equiv 1 \pmod 4$; Seberry and Yamada tabulate order 109 under exactly that mechanism [8, Table A.1, p. 543]. The two routes are therefore the $(q-1)/4$ branch used here and the conference-matrix branch they tabulate. The route through $(q-1)/4 = 27$ is preferred here only because it terminates in an ingredient already admitted by (C.1).) Hence Williamson-type matrices of order 109 exist, and the Turyn array on these together with the T-matrices of order 55 [7, 8, 13, 14] yields a Hadamard matrix of order $4 \cdot 55 \cdot 109 = 23980$. (The Turyn-array statements of [8] are given for Williamson matrices; the Williamson-*type* form used here is [14, Thm. 5].)

The same ingredient is used in the literature in exactly this way: Đoković [30, Prop. 3.4] obtains a Hadamard matrix of order $4 \cdot 79 \cdot 109 = 34444$ from T-sequences of length 79 together with Williamson-type matrices of order 109, citing [8] for the latter. The test of §A.4 misses $q = 109$ because its W-known class admits only two mechanisms — $w \leq 33$, and $2w - 1$ a prime power, which fails here since $2 \cdot 109 - 1 = 217 = 7 \cdot 31$ — whereas Miyamoto's theorem is a third. We therefore leave that test and Table 1 unchanged, $q = 109$ being a genuine failure of the test as stated; as with $q = 73$ in §A.1, the failure is a property of the test, not of Hadamard existence.

A.4 An elementary-witness test and the candidate orders

Beyond the audited range we have no table to appeal to, so we apply a deliberately conservative *elementary-witness* test based on (4.1) and Turyn's array. We call an odd order t *T-known* when $t \leq 59$; the required T-matrices follow from the known base sequences $BS((t+1)/2, (t-1)/2)$, whose second parameter is at most 29. We call an odd order w *W-known* when $w \leq 33$, or when

$2w-1$ is a prime power; in the latter case Turyn's $(P+1)/2$ family, with $P = 2w-1 \equiv 1 \pmod 4$, supplies the required Williamson-type matrices. An unaudited order $N = 4m$ passes the test if N is a Paley I/II order, or if there is an *exhibited* factorization

$$m = tw, \qquad t \text{ } T\text{-known}, \qquad w \text{ } W\text{-known}. \tag{C.1}$$

Thus every passing verdict is an unconditional construction from the ingredients just listed; no multiplicative closure is assumed.

Remark 8. *An earlier version of this appendix asserted that classes of eligible T- and Williamson orders were multiplicatively closed and then classified m prime factor by prime factor. That assertion was uncited, and we withdraw both it and the factorwise test. Even for Williamson matrices proper, unrestricted closure is false: Đoković [9] enumerated Williamson matrices at orders* 33 *and* 39*, but proved that none exist at* $35 = 5{\cdot}7$*, although they exist at both orders* 5 *and* 7*. Holzmann, Kharaghani and Tayfeh-Rezaie [10] later found the further nonexistence orders* 47, 53, *and* 59*. Terminology in the literature includes broader variants called Williamson-type matrices; we make no closure claim for any such class. Instead,* (C.1) *requires the two actual ingredient orders used by Turyn's array to be known individually.*

The factorwise test was unreliable in both directions. It could accept an m merely because each prime factor was eligible, without producing a pair t, w. It could also reject an m because of a prime factor that nevertheless lies inside a directly known composite W-order. For example, at $q = 197$,

$$m = 19503 = 3 \cdot 6501, \qquad 2 \cdot 6501 - 1 = 13001$$

is prime, so $t = 3, w = 6501$ is a witness under (C.1)*; the old factorwise test rejected it because of the prime factor* 197*. This is why the table below is recomputed from actual divisor splits rather than inherited from the earlier screen.*

The audited range of §A.1 is a table lookup, and the $q = 81$ witness of §A.2 cites its two ingredients individually. Both are independent of this correction. Nothing in §§1–4, and in particular neither Theorem 1 nor its proof, depends on this appendix.

A passing verdict now exhibits a construction. A failing verdict (*no restricted elementary witness*) is a statement about this test only, not a claim that N is open. Over the 232 orders with $q \le 3000$, the strict test covers 128 (audited, Paley, or Turyn) and leaves 104 without a witness. The smallest failure remains

$$q = 109, \qquad N = 23980, \qquad m = 5 \cdot 11 \cdot 109,$$

for which none of the possible T-known divisors produces a W-known complementary factor. In particular, $2 \cdot 109 - 1 = 217 = 7 \cdot 31$ is not a prime power. The failure is again a property of the test alone: the single missing ingredient is Williamson-type matrices of order 109, and §A.3 supplies them, so the split $t = 55,\ w = 109$ does cover $N = 23980$ (compare $q = 73$). We do not assert that any entry of Table 1 is open.

Table 1 lists all 104 failures of this restricted test for $q \le 3000$. It is a reproducible classification relative to the stated ingredient sets, not a literature-complete classification of the corresponding Hadamard orders. The ingredient sets themselves are conservative choices rather than the state of the art, and the first entry of the table already illustrates this: §A.3 covers $q = 109$ by a mechanism the test does not admit. Widening the T-known class would remove further entries — [30, §2] records T-sequences of every length $n \le 100$ except possibly $n = 97$, against the bound $t \le 59$ used here. We have not recomputed the table under any widened ingredient set; Table 1 is exactly the output of the rules as stated in (C.1).

Table 1: Orders $N = 2q(q+1)$, q a prime power $\equiv 1 \pmod 4$, with no restricted elementary witness (§A.4), $109 \le q \le 2969$.

q	N	odd part m
109	23980	$5 \cdot 11 \cdot 109$
121	29524	$11^2 \cdot 61$
157	49612	$79 \cdot 157$
173	60204	$3 \cdot 29 \cdot 173$
229	105340	$5 \cdot 23 \cdot 229$
277	154012	$139 \cdot 277$
289	167620	$5 \cdot 17^2 \cdot 29$
313	196564	$157 \cdot 313$
397	316012	$199 \cdot 397$
409	335380	$5 \cdot 41 \cdot 409$
421	355324	$211 \cdot 421$
457	418612	$229 \cdot 457$
529	560740	$5 \cdot 23^2 \cdot 53$
577	667012	$17^2 \cdot 577$
601	723604	$7 \cdot 43 \cdot 601$
613	752764	$307 \cdot 613$
617	762612	$3 \cdot 103 \cdot 617$
641	823044	$3 \cdot 107 \cdot 641$
661	875164	$331 \cdot 661$
733	1076044	$367 \cdot 733$
757	1147612	$379 \cdot 757$
797	1272012	$3 \cdot 7 \cdot 19 \cdot 797$
821	1349724	$3 \cdot 137 \cdot 821$
829	1376140	$5 \cdot 83 \cdot 829$
841	1416244	$29^2 \cdot 421$
853	1456924	$7 \cdot 61 \cdot 853$
877	1540012	$439 \cdot 877$
937	1757812	$7 \cdot 67 \cdot 937$
961	1848964	$13 \cdot 31^2 \cdot 37$
997	1990012	$499 \cdot 997$
1009	2038180	$5 \cdot 101 \cdot 1009$
1021	2086924	$7 \cdot 73 \cdot 1021$
1033	2136244	$11 \cdot 47 \cdot 1033$
1097	2409012	$3^2 \cdot 61 \cdot 1097$
1109	2461980	$3 \cdot 5 \cdot 37 \cdot 1109$
1117	2497612	$13 \cdot 43 \cdot 1117$
1129	2551540	$5 \cdot 113 \cdot 1129$
1153	2661124	$577 \cdot 1153$
1181	2791884	$3 \cdot 197 \cdot 1181$
1201	2887204	$601 \cdot 1201$
1213	2945164	$607 \cdot 1213$
1237	3062812	$619 \cdot 1237$
1249	3122500	$5^4 \cdot 1249$
1297	3367012	$11 \cdot 59 \cdot 1297$
1321	3492724	$661 \cdot 1321$
1361	3707364	$3 \cdot 227 \cdot 1361$
1373	3773004	$3 \cdot 229 \cdot 1373$
1381	3817084	$691 \cdot 1381$
1453	4225324	$727 \cdot 1453$
1481	4389684	$3 \cdot 13 \cdot 19 \cdot 1481$
1553	4826724	$3 \cdot 7 \cdot 37 \cdot 1553$
1597	5104012	$17 \cdot 47 \cdot 1597$
1601	5129604	$3^2 \cdot 89 \cdot 1601$
1609	5180980	$5 \cdot 7 \cdot 23 \cdot 1609$
1637	5362812	$3^2 \cdot 7 \cdot 13 \cdot 1637$

q	N	odd part m
1657	5494612	$829 \cdot 1657$
1681	5654884	$29^2 \cdot 41^2$
1709	5844780	$3^2 \cdot 5 \cdot 19 \cdot 1709$
1753	6149524	$877 \cdot 1753$
1777	6319012	$7 \cdot 127 \cdot 1777$
1789	6404620	$5 \cdot 179 \cdot 1789$
1873	7020004	$937 \cdot 1873$
1877	7050012	$3 \cdot 313 \cdot 1877$
1933	7476844	$967 \cdot 1933$
2017	8140612	$1009 \cdot 2017$
2029	8237740	$5 \cdot 7 \cdot 29 \cdot 2029$
2053	8433724	$13 \cdot 79 \cdot 2053$
2089	8732020	$5 \cdot 11 \cdot 19 \cdot 2089$
2113	8933764	$7 \cdot 151 \cdot 2113$
2137	9137812	$1069 \cdot 2137$
2161	9344164	$23 \cdot 47 \cdot 2161$
2197	9658012	$7 \cdot 13^3 \cdot 157$
2209	9763780	$5 \cdot 13 \cdot 17 \cdot 47^2$
2213	9799164	$3^3 \cdot 41 \cdot 2213$
2269	10301260	$5 \cdot 227 \cdot 2269$
2273	10337604	$3 \cdot 379 \cdot 2273$
2281	10410484	$7 \cdot 163 \cdot 2281$
2297	10557012	$3 \cdot 383 \cdot 2297$
2341	10965244	$1171 \cdot 2341$
2377	11305012	$29 \cdot 41 \cdot 2377$
2381	11343084	$3 \cdot 397 \cdot 2381$
2389	11419420	$5 \cdot 239 \cdot 2389$
2401	11534404	$7^4 \cdot 1201$
2417	11688612	$3 \cdot 13 \cdot 31 \cdot 2417$
2437	11882812	$23 \cdot 53 \cdot 2437$
2473	12236404	$1237 \cdot 2473$
2557	13081612	$1279 \cdot 2557$
2617	13702612	$7 \cdot 11 \cdot 17 \cdot 2617$
2621	13744524	$3 \cdot 19 \cdot 23 \cdot 2621$
2657	14124612	$3 \cdot 443 \cdot 2657$
2689	14466820	$5 \cdot 269 \cdot 2689$
2693	14509884	$3 \cdot 449 \cdot 2693$
2713	14726164	$23 \cdot 59 \cdot 2713$
2749	15119500	$5^3 \cdot 11 \cdot 2749$
2777	15429012	$3 \cdot 463 \cdot 2777$
2797	15652012	$1399 \cdot 2797$
2801	15696804	$3 \cdot 467 \cdot 2801$
2833	16057444	$13 \cdot 109 \cdot 2833$
2837	16102812	$3 \cdot 11 \cdot 43 \cdot 2837$
2857	16330612	$1429 \cdot 2857$
2909	16930380	$3 \cdot 5 \cdot 97 \cdot 2909$
2917	17023612	$1459 \cdot 2917$
2953	17446324	$7 \cdot 211 \cdot 2953$
2969	17635860	$3^3 \cdot 5 \cdot 11 \cdot 2969$

Every odd part in Table 1 is composite by (4.1), so these orders do not meet the $4 \times$ prime form that dominates the initial list of open Hadamard orders. At the larger sizes in the table we make no literature-complete assertion of known or open status. Widening the ingredient set (for example with computer-tabulated Williamson and T-matrices or Goethals–Seidel cyclotomic difference sets) can remove entries from this restricted list. The role of Theorem 1 is to give a single closed-form rule for the whole family, including the region this elementary screen does not reach, rather than to derive novelty from a coverage-table gap.

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